

Ex.No.4: OFDMA Resource Block Allocation Date

Objective 1 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
% System Parameters
```

```
bandwidth = 10e6;    % 10 MHz
```

```
rb_size = 180e3;     % 180 kHz per Resource Block
```

```
% Total Resource Blocks
```

```
total_RB = floor(bandwidth / rb_size);
```

```
% Users
```

```
users = 1:6;
```

```
% Resource Block Allocation
```

```
base = floor(total_RB / length(users));
```

```
alloc = base * ones(1, length(users));
```

```
% Distribute Remaining Resource Blocks
```

```
alloc(1:mod(total_RB, length(users))) = ...
```

```
    alloc(1:mod(total_RB, length(users))) + 1;
```

% Resource Block Utilization

utilization = (alloc / total_RB) * 100;

% Display Results

disp('Total Resource Blocks:');

disp(total_RB);

disp('RB Allocation per User:');

disp(alloc);

% Plot Graphs

figure;

% Graph 1: Resource Block Allocation

subplot(2,1,1);

stem(users, alloc, 'filled', 'LineWidth', 2);

title('OFDMA Resource Block Allocation per User');

xlabel('User Index');

ylabel('Number of Resource Blocks');

grid on;

% Graph 2: Resource Block Utilization

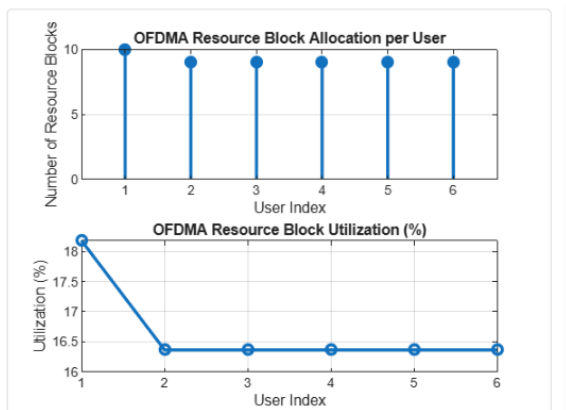
subplot(2,1,2);

plot(users, utilization, '-o', 'LineWidth', 2);

```

title('OFDMA Resource Block Utilization (%)');
xlabel('User Index');
ylabel('Utilization (%)');
grid on;

```



Objective 2 Matlab Code and Output:

```

clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];

% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')
disp(util)

```

figure

% ----- Graph 1: RB Allocation -----

```
subplot(2,1,1)
```

```
plot(users, alloc, '-o', 'LineWidth', 2)
```

```
title('OFDMA Resource Block Allocation per User')
```

```
xlabel('User Index')
```

```
ylabel('Number of Resource Blocks')
```

```
grid on
```

% ----- Graph 2: Utilization (%) -----

```
subplot(2,1,2)
```

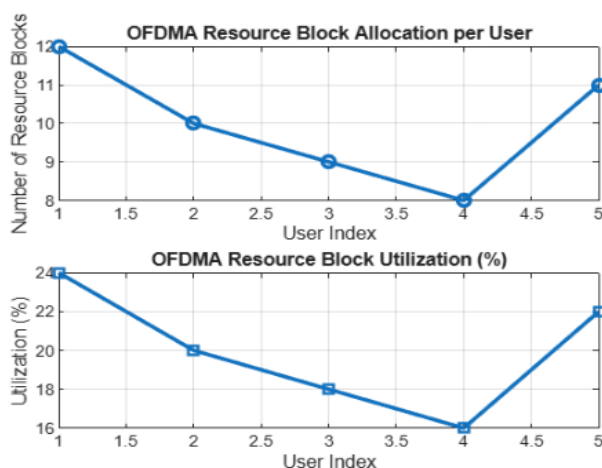
```
plot(users, util, '-s', 'LineWidth', 2)
```

```
title('OFDMA Resource Block Utilization (%)')
```

```
xlabel('User Index')
```

```
ylabel('Utilization (%)')
```

```
grid on
```



Objective 3 Matlab Code and Output:

```
clc; clear; close all;

% System parameters

rb_bw = 180e3; % Bandwidth per RB (180 kHz)


users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')

disp(throughput)

figure

% ----- Graph 1: Throughput per User -----

subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on
```

% ----- Graph 2: RB vs Throughput -----

```
subplot(2,1,2)
```

```
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
```

```
title('Resource Blocks vs Throughput Relationship')
```

```
xlabel('Allocated Resource Blocks')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

4.5200
7.2000
9.7200
5.7600
3.9600

